

DRMD: Explainable Depression Detection Based on Metaphorical Conceptual Mapping

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Abstract

Metaphors are a fundamental cognitive tool for articulating abstract and subjective experiences and implicit semantics, making them potent indicators of psychological state, particularly in individuals with depression. The proliferation of social media has created a vast repository of such metaphorical expressions, offering an unprecedented opportunity to understand mental health struggles. These metaphors can provide crucial insights for clinical assessment and therapeutic intervention. However, their potential remains largely untapped in automated depression detection, primarily due to the lack of large-scale, annotated datasets. To bridge this gap, we introduce the Depression-Related Metaphor Dataset (DRMD), a novel resource of social media posts related to depression, incorporating depression levels (severe, moderate, minimum, and null), the presence or absence of metaphors, and their conceptual source domain mappings. We leverage this dataset to fine-tune Large Language Models (LLMs), integrating metaphorical features to enhance detection capabilities. Our results demonstrate that models incorporating metaphorical information achieve superior accuracy in depression detection and, importantly, generate high-quality explanations for their decisions by referencing specific metaphorical expressions. This work underscores the critical role of metaphorical analysis in computational mental health and provides a foundation for future research in explainable AI for depression detection. The dataset is publicly available.¹

CCS Concepts

• **Applied computing** → *Document management and text processing*.

Keywords

Depression Detection, Social Media, Implicit Semantics, Large Language Models, Mental Health Diagnostics, Metaphor Analysis

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¹<https://github.com/xinyu706/DRMD>



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1 Introduction

Depression represents a pervasive global health challenge, yet its detection remains fraught with obstacles. Conventional diagnostic methods, which rely heavily on clinical interviews and self-reporting, can be compromised by social stigma, limited access to healthcare, and a patient's own lack of insight [13, 34, 40]. In response to these limitations, computational mental health research has turned to social media as a source of non-intrusive, naturalistic data [1, 3, 10]. These digital platforms offer a unique window into personal expression, where users often articulate their innermost thoughts and emotional states.

A particularly revealing dimension of this self-expression is the use of metaphorical language. Grounded in cognitive linguistics, [16] seminal work posits that metaphors are fundamental structures that shape human thought and communication, allowing us to conceptualize abstract experiences and implicit semantics in terms of more concrete, tangible domains. For individuals grappling with depression, metaphor becomes an essential tool for structuring and conveying overwhelming subjective experiences [5, 21, 28]. Expressions such as "being trapped in a dark hole," "carrying a heavy weight," or "fighting a losing battle" provide rich, qualitative insight into a person's psychological reality—insight that often eludes literal description. Consequently, metaphorical expressions can serve as potent indicators of underlying cognitive and emotional processes, holding significant value for clinical assessment.

Despite this potential, the field of automated depression detection has largely overlooked the systematic analysis of metaphorical language. Prevailing computational approaches predominantly leverage lexical, syntactic, and sentiment-based features [9, 12, 17, 30, 36, 39, 49], which, while useful, often fail to capture the nuanced conceptual frameworks that metaphors encode. The primary impediment to integrating metaphorical analysis has been a critical resource gap: the scarcity of large-scale, annotated datasets that specifically label depression-related metaphors and their underlying conceptual mappings. While some metaphor-annotated corpora

exist [5, 14, 20, 35], none are tailored to the context of depression analysis. Without such dedicated resources, it is challenging to train computational models to reliably recognize and interpret these critical linguistic signals, leaving a substantial and informative layer of data untapped.

To address this gap, this paper introduces the Depression-Related Metaphor Dataset (DRMD), a novel, publicly available resource comprising social media posts from individuals with depression, each annotated for metaphorical content and its corresponding conceptual source domain. We demonstrate the utility of this dataset by leveraging it to fine-tune Large Language Models (LLMs), effectively integrating metaphorical awareness into their architecture. Our research investigates the central hypothesis that incorporating metaphorical features can significantly enhance the performance and explainability of automated depression detection systems. The contributions of this work are threefold:

- (1) We introduce the first large-scale, fine-grained dataset that annotates social media posts with depression severity levels, the presence and type of metaphors, and their corresponding conceptual source domains. This resource provides a unique foundation for metaphor-aware mental health analysis, enabling future research into the semantic and cognitive markers of depression in online discourse.
- (2) We propose a novel fine-tuning methodology for large language models (LLMs) that integrates metaphorical features and conceptual source domain mappings. This approach enhances depression detection performance and enables models to generate interpretable, conceptually grounded explanations for their predictions. Experimental results demonstrate the critical role of metaphor understanding in improving both accuracy and explainability in depression detection.
- (3) We construct a diversified instruction-tuning dataset derived from our annotated resource to fine-tune two baseline LLMs. The metaphor-enhanced models exhibit not only superior depression detection capability but also the ability to produce human-intelligible explanations for their reasoning. These results represent a concrete step forward toward explainable AI in the context of mental health analytics.

2 Related Work

2.1 Depression Detection Datasets

Mental health datasets vary in task design, data sources, and annotation scope. Most publicly available resources focus on binary or multi-class depression detection [6, 23, 25, 26, 32, 42], while a smaller portion simulates counseling dialogs [15, 37]. Social media platforms such as Reddit, Twitter, and Weibo remain primary data sources, yielding large-scale corpora like MDDL [32] (292,000 tweets) and Reddit-based datasets [42]. In contrast, counseling session datasets (e.g., D4 [45]) are smaller but offer richer emotional context.

Most existing datasets provide only depression labels or user-level annotations, rarely capturing metaphorical or conceptual linguistic features, and they typically focus on literal text interpretation. Consequently, computational models trained on these datasets often overlook the cognitive and affective nuances embedded in non-literal expressions, limiting interpretability.

2.2 Depression metaphor

Recent metaphor-oriented studies have highlighted how individuals convey mental health experiences through figurative language. For example, [41] examined metaphorical expressions in online mental health forums using Conceptual Metaphor Theory and Self-Determination Theory, emphasizing depression narratives. However, this analysis remains largely qualitative, without annotated datasets or severity labels suitable for computational modeling.

Similarly, [5] introduced The Mental Health Metaphor Dictionary, a Spanish-language repository of conceptual metaphors from first-person accounts of individuals with severe mental disorders, including schizophrenia, bipolar disorder, depression, and obsessive-compulsive disorder. While this dictionary provides a rich domain-specific resource, it is not depression-specific, lacks sample-level depression annotations or severity labels, and is primarily intended for linguistic or educational purposes rather than computational modeling.

By contrast, the Depression-Related Metaphor Dataset (DRMD) we propose is English-language, depression-focused, and designed for computational applications. It integrates five complementary tasks—depression detection, severity classification, metaphor identification, source-domain mapping, and explanation generation—effectively bridging metaphor theory with task-oriented, explainable depression modeling.

3 Dataset

Problem Statement: Existing social media datasets for depression analysis primarily focus on binary or multi-class depression labels while largely overlooking the role of metaphorical language. Metaphors can provide important cues for depression detection and analysis, helping to reveal its cognitive and emotional manifestations. To address this gap, we aim to construct a multi-layered dataset containing depression labels, depression severity labels, metaphor labels, and depression-related conceptual source domain labels. This dataset is designed to support research on explainable depression detection and to facilitate the exploration of metaphor usage patterns in depressive texts.

In this section, we describe the dataset construction pipeline process. We integrated five publicly available depression detection datasets and employed large language models (LLMs) to automatically annotate them with metaphor labels and depression-related conceptual source domain labels. After manual correction, the finalized dataset was obtained. The full dataset construction pipeline—including data collection and filtering (clean and unite) as well as annotation (metaphor annotation and depression-related conceptual source domain annotation)—is illustrated in Figure 1.

3.1 Data Collection and Filtering

We began by gathering publicly accessible depression-related datasets that included both binary depression detection and multi-class depression severity classification. Based on their availability, post completeness, and clarity of annotation, we selected five datasets and summarized their label types and sample sizes in Table 1. To ensure data quality and consistency, we conducted a two-step pre-processing pipeline:

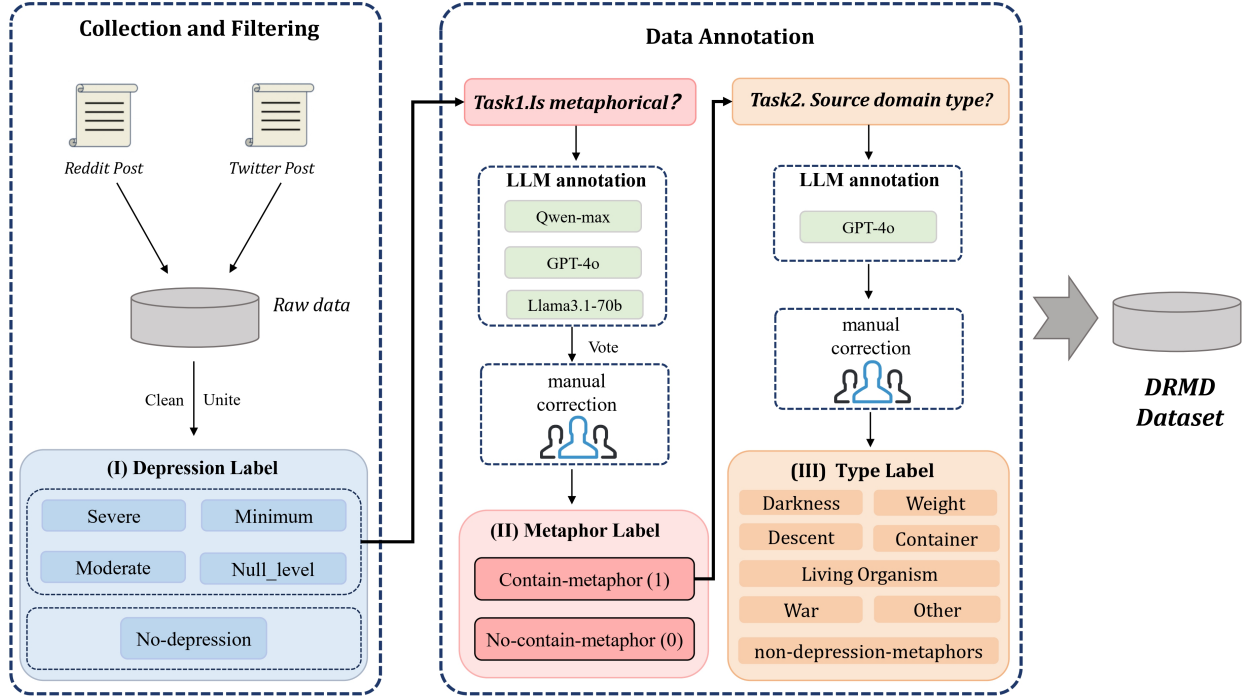


Figure 1: Construction pipeline of Depression-Related Metaphor Dataset (DRMD): (1) Data collection and filtering; (2) Metaphor annotation via multi-LLM voting and manual correction; (3) Source domain annotation for depression-related conceptual mappings;

Data Cleaning: We removed duplicate entries, meaningless phrases, and noisy samples, and filtered out special characters using regular expressions. Since large language models (LLMs) struggle to capture information from the middle of overly long texts [19], we further applied length-based filtering by excluding samples shorter than 5 characters or longer than 7,000 characters.

Label Unification: To integrate the cleaned binary and multi-class datasets, we defined two label fields: 'label' (for depression detection) and 'level' (for depression severity detection). Binary datasets were standardized into two categories: 'depression' and 'no-depression'. For multi-class severity datasets, all depression-related samples were mapped to the 'depression' category while preserving their severity levels. To facilitate downstream instruction construction described in Section 5.1, we labeled samples from binary datasets as 'depression_null_level' and those from multi-class datasets as 'depression'. Therefore, the complete set of depression samples includes both 'depression_null_level' and 'depression' categories. Specifically, in the HelaDepDet dataset [23], the original four severity levels ('severe', 'mild', 'minimum', 'moderate') were merged into three categories by combining 'mild' with 'minimum'. This decision was motivated by the high semantic overlap and similar symptom intensity between the 'mild' and 'minimum' categories. The unified severity-labeled samples were then merged with those from the Kayalvizhi and Thenmozhi dataset [29], and the remaining 'no-depression' samples were added to the binary group. Finally,

to maintain an approximately balanced 1:1 ratio between depression and non-depression samples, we performed random sampling. The resulting corpus contains 10,473 'depression' samples, 1,663 'depression_null_level' samples, and 12,130 'no-depression' samples, forming the foundation for subsequent LLM-based automatic annotation and manual verification in depression detection and severity assessment tasks.

Table 1: Overview of the original datasets, including platforms, label types, and sample counts.

Dataset	Platform	Label	Samples
HelaDepDet [23]	Reddit	Severity	41,819
Kayalvizhi and Thenmozhi [29]	Reddit	Severity	10,352
Pirina and Çöltekin [26]	Reddit	Binary	3,009
DDRC [11]	Reddit	Binary	7,729
MDDL [32]	Twitter	Binary	11,875
Total	-	-	74,784

3.2 Metaphor Annotation

We conducted metaphor annotation as the first task, framing it as a binary labeling problem to determine whether each depression-related sample contains metaphorical expressions. Since our analysis focuses on metaphors used by individuals with depressive

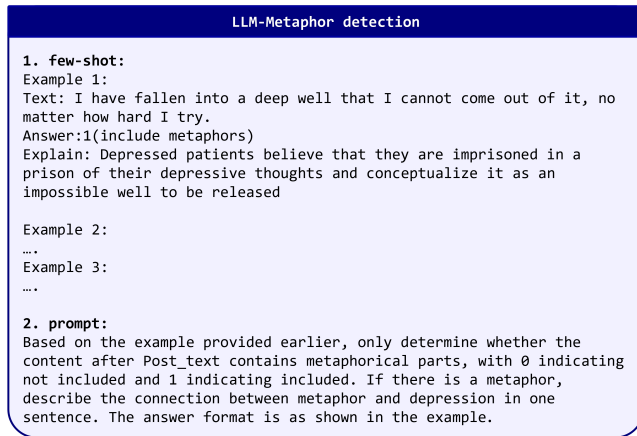


Figure 2: Prompt used for metaphor annotation, including few-shot examples and the instruction template.

tendencies, we restricted annotation to samples labeled as 'depression' or 'depression_null_level'.

We combined large language model (LLM)-based automatic annotation with manual correction to identify metaphorical expressions in the cleaned and unified depression dataset. Leveraging the strong capabilities of LLM in understanding figurative and context-dependent language [4, 22, 31, 43, 44], We combined large language model (LLM)-based automatic annotation with manual correction to identify metaphorical expressions in the cleaned and unified depression dataset. Leveraging the strong capabilities of LLMs in understanding figurative and context-dependent language [4, 22, 31, 43, 44], we employed three state-of-the-art models—Qwen-Max, LLaMA-70B, and GPT-4—for automatic annotation. As illustrated in Figure 2, each annotation prompt consisted of two components: a set of [few-shot] examples designed to guide the models toward consistent metaphor identification and a [prompt] instruction specifying the annotation criteria and output format.

After obtaining the model-generated annotations, we retained only those samples for which all three models produced consistent outputs (i.e., all labeled as contain-metaphor or all labeled as no-contain-metaphor), ensuring high-confidence labels. For samples with conflicting outputs, manual verification and correction were conducted to resolve disagreements. This hybrid strategy allowed us to efficiently balance annotation quality and scalability. Among the 12, 136 depression-related samples, 6, 025 were labeled as contain-metaphor (further used in source-domain annotation), and 6, 111 as no-contain-metaphor, constituting the final dataset for the metaphor detection task.

3.3 Depression-Related Conceptual Source Domain Annotation

The second annotation task focuses on identifying depression-related conceptual source domains and follows the metaphor annotation step. We applied this task only to samples labeled as 'contain-metaphor' in the previous stage, as our goal was to label conceptual source domains relevant to depression.

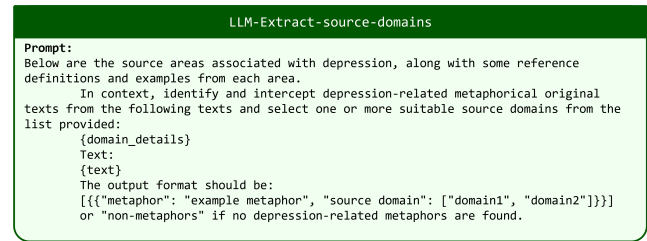


Figure 3: Prompt used for conceptual source domain annotation. Definitions and examples for seven source domains plus 'Other' guide GPT-4 in accurate selection.

Drawing on prior research on depressive metaphors and an established source domain taxonomy [2, 5, 21, 38, 41], together with our analysis of the extracted metaphor data, we grouped depression-related metaphors into six primary conceptual source domains. We also defined an additional 'Other' category to capture any remaining cases. Table 7 presents the classification of metaphorical source domains related to depression and the 'Other' category, along with their definitions and representative examples. This taxonomy provides a conceptual foundation for understanding how depressive experiences are linguistically structured through metaphors. It also serves as a guideline for dataset annotation and model training in the tasks of source domain identification and metaphor explanation. By explicitly defining each source domain (e.g., Darkness, Weight, Container, War), the table ensures semantic consistency across data labeling and facilitates an explainable analysis of model outputs.

We employed GPT-4 to automatically identify the corresponding source domain for each metaphorical sample. Specifically, GPT-4 was tasked with locating the depression-related conceptual source domain within each metaphorical expression and selecting from the six predefined domains. If a metaphor mapped to one or more domains, it was assigned accordingly. Metaphors that could not be mapped to any predefined domain but remained depression-related were labeled as 'Other'. To ensure precise annotation, GPT-4 returned the metaphorical sentence corresponding to the selected source domain. If a sample contained no depression-related conceptual source domain (i.e., the metaphor was unrelated to depression), the model returned a statement indicating the absence of such mapping.

To guide GPT-4 for accurate selection, we provided definitions and examples for all seven source domains, as illustrated in Figure 3. We also used standardized prompt templates to regulate model responses and facilitate subsequent manual verification. After manual correction, the final dataset includes 4, 426 depression-related conceptual source domain samples and 1, 599 non-depression-related metaphor samples, supporting the task of depression-related conceptual source domain identification.

3.4 Manual Evaluation

The manual evaluation team consisted of six members, all native English speakers with a background in cognitive linguistics. To ensure high-quality annotations, we developed detailed correction guidelines and documentation, including instructions, examples, and key considerations for both metaphor annotation and

Table 2: Distribution of dataset labels across different tasks.

Task	Detailed Labels	Number
Depression detection (label)	depression	10,473
	depression_null_level	1,663
	no-depression	12, 136
Depression severity detection (level)	minimum	3,615
	moderate	3,541
	severe	3,317
Metaphor detection (metaphor)	contain-metaphor	6, 025
	no-contain-metaphor	6, 111
Conceptual metaphor source domains (source)	metaphor source domain related to depression	4,426
	non-depression metaphors	1,599

depression-related conceptual source domain annotation. We conducted training sessions before each correction round and continuously updated the materials to address emerging issues.

Manual correction included two main components: **(1) Binary correction of metaphorical expressions:** Each team member independently selected the most accurate annotation from the outputs of the three candidate LLMs, following the training guidelines. The final label was determined by majority voting. If votes were tied, the team discussed collaboratively to reach a consensus. **(2) Correction of depression-related conceptual source domains:** The six members were randomly divided into three groups. Two groups independently corrected the model-predicted source domains. When disagreements arose, the third group adjudicated. This task included two subtasks: (a) verifying whether the model-predicted depression-related source domain matched the metaphor. If it did not, all three groups discussed and selected the correct source domain and extracted the corresponding metaphorical sentence to enhance annotation accuracy; (b) verifying samples where the model predicted no depression-related source domain. Misclassified samples were reassigned to the correct source domain, with the corresponding sentence extracted. Note that this evaluation focused solely on model-generated outputs, not on metaphors that the model failed to identify.

To assess annotation consistency, we randomly selected 200 samples for each task. The Fleiss' Kappa coefficient [7] reached $k \geq 0.8$ for both tasks, demonstrating high reliability for both metaphor and source domain corrections.

After LLM-based automatic annotation and manual correction, Figure 4 presents examples from the final annotated dataset.

4 Dataset Analysis

The final unified dataset, completed after metaphor annotation and depression-related conceptual source domain annotation, supports four tasks, each corresponding to a distinct annotation level: (1) **Depression detection** (label): binary classification of depression vs. non-depression; (2) **Depression severity detection** (level): three-class classification (minimum, moderate, severe); (3) **Metaphor detection** (metaphor): binary labeling of metaphor presence; and (4) **Depression-related conceptual metaphor source domain identification** (source): classification of depression-related vs.

```
{
  "text": "Time to get biblical (rant: as a person who battles with demons everyday, I want to believe that in the end, our actions define us, not our urges or desires",
  "label": "depression",
  "level": "moderate",
  "metaphor": "contain-metaphor",
  "source": {"metaphor": "as a person who battles with demons everyday.", "source domain": ["War", "Living Organism"]}
}
```

Figure 4: Example of a manually corrected metaphor annotation. Each post may contain one or more labels, ranging from the minimal 'label' (depression detection) to 'label' and 'level' (depression severity detection), as well as 'metaphor' (metaphor detection) and 'source' (conceptual metaphor source domain identification).

non-depression-related conceptual metaphor source domains. Table 2 summarizes the label distribution for each task.

4.1 Depression Metaphor

As shown in Table 2, 6, 025 samples from users with depressive tendencies contain metaphors, representing approximately 50% of the depression-related data. This proportion is substantially higher than the average metaphor usage in general language (about 32%) [33], indicating that individuals with depressive tendencies tend to rely more on metaphors to express complex internal experiences. Among these samples, the majority (73%) map to the six predefined depression-related conceptual source domains or to the 'Other' depression-related category, further validating the robustness of our source domain classification.

4.2 Source Domain Categories

Figure 5 illustrates the distribution of the six major depression-related conceptual metaphor source domains, along with the 'Other' category. The most frequent domain is 'Descent', reflecting the core experience of emotional decline in depression. The second most common domain is 'Living Organism', in which depression is metaphorically represented as a harmful or aggressive living entity (e.g., demons or monsters). The 'Container' domain appears

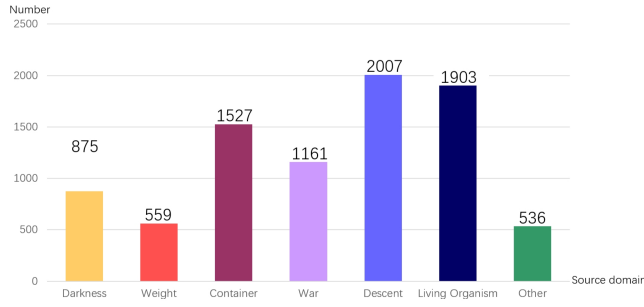


Figure 5: Distribution of source domains for depression-related conceptual metaphors.

over 1,500 times, highlighting the feeling of emotional confinement often reported by individuals with depression.

Excluding the 'Other' category, the 'Weight' domain is the least frequent, suggesting that depression is primarily conceptualized as a psychological rather than a physiological experience, although some individuals do report physical symptoms. Common metaphors in the 'Other' category include conceptualizations of depression as a journey or as a ticking time bomb in the mind. These distribution patterns are consistent with prior studies on depression-related metaphors.

5 Experiment

5.1 Instruction Construction

To enable explainable depression detection based on metaphorical concept mapping, we designed a systematic instruction set construction framework integrated with multi-task fine-tuning. We introduced four types of labels corresponding to core tasks: Depression detection, Depression severity detection, Metaphor detection (metaphor), and Depression-related conceptual metaphor source domain identification (source). These labels support individual classification tasks and collectively enable a fifth task—metaphor-driven depression analysis—where the model generates interpretable analyses of depressive content by detecting metaphors and analyzing the meanings of their depression-related source domains. This design allows the model to acquire task-specific knowledge while leveraging metaphorical cues to produce explainable depression insights.

To enhance explainability and generalization, we adopted a five-task instruction-tuning framework: a hybrid selection strategy for instruction construction. Specifically, the dataset was partitioned into seven subsets based on label types, and for each subset, seven analysis-depth templates were designed, covering five tasks, including metaphor-driven depression analysis. Task templates were initially defined according to task objectives, label definitions, and prior metaphor analysis studies, and then augmented with GPT-generated linguistic variants to increase diversity. Each set contained five semantically similar but differently worded instruction prompts.

For each dataset detail label, five semantically equivalent label text descriptions were manually generated. For instance, in the depression detection task, the label 'depression' was expanded

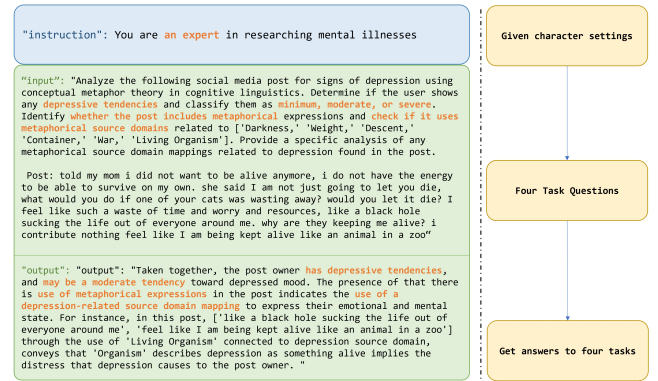


Figure 6: Example of the constructed instruction set. The orange section in the output is replaced according to different labels.

into variants such as "has depressive tendencies" and "is showing signs of depression." During instruction generation, one variant was randomly selected for each label and inserted into the corresponding pre-defined template. This strategy exposes the model to diverse linguistic forms while maintaining semantic consistency, enabling robust learning of label meanings across contexts. Finally, the model was fine-tuned on the union of instruction sets from all seven subsets. Figure 6 presents an example instruction.

This instruction design enables the model to capture cognitive and emotional cues conveyed through metaphorical language in depression-related posts. In the fifth task, the model integrates metaphor detection and source domain identification to generate interpretable analyses of depressive content, including potential depressive tendencies and, when available, severity levels, this approach enhances the model's performance on depression detection, depression severity detection, metaphor detection, and depression-related conceptual metaphor source domain identification, while providing a structured framework for interpretable, metaphor-informed depression analysis.

5.2 Experimental Setups

We used Qwen2.5-7B [27] and LLaMA 3.1-8B [8] as baseline models. After fine-tuning, both models adapted effectively to domain-specific tasks, achieving high performance with relatively low computational cost. Training was conducted on two NVIDIA 4090 24GB GPUs, using a learning rate of 1×10^{-4} , 16-bit half-precision floating-point precision, and gradient accumulation steps of 8. Fine-tuning leveraged the LLaMA Factory optimization framework [48].

5.3 Results

5.3.1 Correctness. We evaluated model performance using the Macro-F1 score, accuracy, and recall across three settings: Zero-Shot (ZS), Few-Shot (FS), and Fine-Tuned (FT). As shown in Table 3, fine-tuned models generally achieved notable improvements across all tasks. In particular, LLaMA-3.1-FT exhibited an approximately 10% increase in the Macro-F1 score for depression detection and a 27% improvement for depression severity detection. In the binary-classification metaphor detection task (Table 3), LLaMA-3.1-FT and

Table 3: Performance of different models (%) across depression detection, depression level recognition, and metaphor detection tasks. "ZS" denotes zero-shot, "FS" denotes few-shot, and "FT" denotes fine-tuning.

Model	Parameter	Depression detection			Depression level recognition			Metaphor detection		
		Macro-F1	Accuracy	Recall	Macro-F1	Accuracy	Recall	Macro-F1	Accuracy	Recall
LLaMA3.1-ZS	8B	87.93	81.41	85.59	37.85	39.32	39.32	65.65	70.25	57.19
LLaMA3.1-FS		75.78	63.02	91.51	31.43	37.78	37.78	72.92	72.72	73.88
LLaMA3.1-FT		97.03	97.04	96.55	64.86	64.32	64.32	97.57	97.62	96.36
Qwen2.5-ZS	7B	78.84	81.74	67.95	46.23	53.38	53.38	83.13	80.53	96.53
Qwen2.5-FS		72.82	68.04	85.54	48.43	50.00	50.00	85.37	83.81	95.04
Qwen2.5-FT		82.49	84.99	70.67	51.93	60.25	60.25	97.15	97.21	95.87
Gemma-ZS		74.80	74.32	77.17	44.47	44.78	44.78	41.46	45.19	99.36
Gemma-FS		66.41	49.77	99.59	30.39	32.47	32.47	49.09	58.27	99.59
Mistral-ZS		79.42	79.98	77.32	43.47	44.97	44.97	51.34	63.98	97.83
Mistral-FS		67.01	52.45	96.86	32.17	33.73	33.73	38.79	34.47	100.00
LLaMA3.1-70B-ZS	70B	79.20	81.75	70.35	52.41	52.00	52.00	72.01	85.36	93.47
LLaMA3.1-70B-FS		78.85	79.80	75.50	34.02	34.41	34.41	71.05	83.87	98.16
GPT-4o-ZS	1800B	77.73	81.20	67.77	56.89	58.40	58.40	63.52	76.80	99.02

Table 4: Performance of different models in a source domain identification task.

Model	Source Domain Identification		
	BLEU-4	ROUGE-1	BertScore
LLaMA3.1-ZS	19.75	22.43	81.74
LLaMA3.1-FS	19.27	28.11	81.66
LLaMA3.1-FT	23.26	46.01	88.35
Qwen2.5-ZS	13.28	23.71	81.11
Qwen2.5-FS	20.09	27.71	80.39
Qwen2.5-FT	31.5	43.67	88.05

Qwen2.5-FT achieved Macro-F1 score gains of roughly 32% and 14%, respectively, demonstrating the effectiveness of our dataset and five-task instruction-tuning framework.

We further assessed the models' ability to identify depression-related source domains. Given that model outputs may include descriptive statements rather than exact source domain labels, we employed semantic similarity metrics for evaluation (Table 4). Results indicate that fine-tuned models generally outperform baseline models in identifying depression-related source domains, though there remains room for improvement. Refining the classification schema and providing clearer definitions for these source domains may enhance future performance.

5.3.2 Evaluation. We used multiple automated metrics to assess the quality of model-generated explanations (Table 5). BLEU-4 [24] measures exact 4-gram matches between generated and reference outputs. ROUGE-L [18] evaluates semantic consistency at the sentence or paragraph level. BERTScore [46] captures deeper semantic similarity, often providing more precise evaluation results than n-gram-based metrics. Inspired by MT-Bench [47], we also employed LLaMA-3.1-70B to score generation quality from three aspects: correctness, rationality, and conciseness, with each aspect scored from 0 to 3. Correctness evaluates alignment with reference answers,

rationality examines logical coherence, and conciseness assesses redundancy or repetition. The final score was obtained by averaging these three dimensions.

We analyzed the average token length of model responses. Fine-tuned Qwen2.5-FT generally produced shorter responses closer to reference answers, whereas baseline models often generated lengthy outputs containing irrelevant content or selected inappropriate conceptual source domains for metaphors, resulting in lower evaluation scores. GPT-4o showed mixed results: in zero-shot settings, it tended to generate extensive analyses that, although often reaching correct conclusions, diverged significantly from reference answers, leading to lower BLEU-4 and ROUGE-L scores while performing better on BERTScore and MT-Bench, these evaluations demonstrate that our fine-tuning strategy, leveraging DRMD and metaphorical source-domain annotations, enhances not only the predictive accuracy of depression detection but also the interpretability and conciseness of model-generated explanations. By employing a five-task instruction-tuning framework to explicitly integrate metaphorical knowledge, models are better able to ground their explanations in human-understandable cognitive structures, a crucial step toward explainable computational mental health systems.

5.4 Supplementary Experiment

To investigate the contribution of annotated metaphorical information in enhancing LLM performance on depression detection, we designed a supplementary experiment. Although fine-tuned Qwen2.5-7B exhibits lower overall performance than LLaMA-3.1-8B in depression detection, the baseline LLaMA-3.1-8B demonstrates instability. Specifically, its outputs often include unavailable responses, which refer to cases where the model refuses to perform the task and returns generic rejection statements (e.g., "I cannot answer this question" or similar safety-related responses) instead of providing an analysis of depressive tendencies or mental health status. Such cases reduce performance consistency and may also

Table 5: Evaluation of model generation results.

Model	BLEU-4	ROUGE-L	BertScore	MT-bench	Average token length
LLaMA3.1-ZS	18.93	10.78	82.83	1.54	111.45
LLaMA3.1-FT	38.44	35.71	90.52	2.10	25.51
Qwen2.5-ZS	10.58	7.81	82.01	1.66	312.00
Qwen2.5-FT	44.80	39.90	91.05	2.33	35.27
GPT-4o-ZS	0.62	13.61	84.70	1.94	205.00

Table 6: Performance after constructing the instruction set fine-tuning model using different labels, with the labels used in constructing the instruction set in parentheses.

Model	Depression detection			Depression level recognition		
	Macro-F1	Accuracy	Recall	Macro-F1	Accuracy	Recall
Qwen2.5	72.81	70.75	78.23	36.36	29.65	29.65
Qwen2.5-FT						
+ (metaphor)	81.87	81.32	84.22	74.96	73.02	73.02
+ (metaphor + source)	97.35	97.41	95.23	74.25	72.45	72.45
+ (metaphor + source + source_location)	97.99	98.03	96.38	74.43	72.45	72.45

occur in other tasks. The proportion of unavailable outputs across tasks is reported in Appendix A.2, reflecting model reliability and adaptability.

Based on these observations, we conducted the supplementary experiment using Qwen2.5-7B. We fine-tuned the model with instruction sets constructed using three types of labels and evaluated its performance on two tasks: depression detection and depression severity detection. Results (Table 6) indicate that incorporating metaphor information improves the model’s ability in depression detection compared with the baseline. Adding metaphor source domain information further enhances performance, while including metaphor locating sentences yields only marginal gains. In the depression severity detection task, metaphor information has a limited impact, and differences among the three fine-tuned models are minor. We hypothesize that this is because metaphors employed by individuals with varying degrees of depression are largely similar, and there is no evidence suggesting substantial differences in the conceptual source domains they map to.

6 Conclusion

In this work, we introduce the Depression-Related Metaphor Dataset (DRMD), a publicly available resource comprising social media posts annotated with fine-grained depression levels, the presence of metaphorical expressions, and their corresponding conceptual source domains. Leveraging this dataset, we designed a five-task instruction-tuning framework to integrate metaphorical and conceptual mapping knowledge into Large Language Models (LLMs). Experimental results demonstrate that models fine-tuned with our framework achieve significant improvements across multiple tasks, including depression detection, metaphor identification, and explanation generation. Moreover, automated evaluation metrics indicate that these models produce more concise, coherent, and interpretable explanations by referencing specific metaphorical expressions, highlighting the value of incorporating cognitive-linguistic features in

computational mental health applications. This study underscores the importance of metaphorical analysis for both the predictive accuracy and interpretability of automated depression detection systems. In the future, we plan to enhance the dataset by refining source-domain annotations, extend the framework to other mental health conditions, and explore the integration of metaphorical information across different languages and cultural contexts.

Ethics and Societal Impact

The DRMD dataset integrates and re-annotates multiple publicly available social media depression datasets collected in compliance with platform terms of service (e.g., Reddit and Twitter). All data were further processed with additional manual and automated de-identification to remove usernames, identifiable references, and sensitive personal narratives.

This study involves secondary analysis of de-identified, publicly accessible research data and is exempt from full institutional review, while adhering to ethical standards for mental health research.

All labels in DRMD are non-clinical. Metaphor annotations and depression-related conceptual source domain annotations are generated through LLM-assisted annotation with expert linguistic verification and are intended solely for research on metaphorical representation and NLP-based mental health indicators, not for clinical diagnosis or real-time monitoring.

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A Appendix

A.1 Source Domain Definitions Detail

Table 7 presents the definitions and representative examples of the seven metaphorical source domains used in our study. Each domain captures a distinct conceptualization of depression: Darkness emphasizes dark or gloomy states; Weight conveys heaviness of body or spirit; Container reflects bounded or constrained experiences; War frames depression as an antagonistic force; Descent represents a downward progression of mood or state; Living Organism portrays depression as a living entity or biological phenomenon; and Other includes metaphors that do not fit the six primary domains, such as abstract comparisons or conceptual journeys. This taxonomy provides a structured foundation for both dataset annotation and the training of models for depression-related metaphor identification, ensuring semantic consistency and facilitating interpretable analysis of model outputs.

A.2 Unavailable Responses

The proportions of unavailable responses across different models and tasks, as shown in Table 8, these proportions vary substantially across models, tasks, and experimental settings. For instance, baseline LLaMA-3.1 in few-shot settings shows extremely high False Ratios in depression level recognition (95.71%), indicating that limited task-specific examples are insufficient for stable predictions. In contrast, fine-tuned models such as Qwen2.5-FT-ZS exhibit much lower proportions of unavailable responses (1.42%), reflecting improved adaptation to the task. Models like gemma-ZS demonstrate strong zero-shot capabilities in metaphor detection, with only 1.27% unavailable responses, highlighting their ability to generalize without task-specific fine-tuning.

Table 7: Source Domain Definitions and Examples.

Source domain	Definition	Example
Darkness	This source domain portrays depression as a state conceptually dominated by dark colors.	1.A gray cloud is crying inside me. 2.It feels like a weight is on me at all times.
Weight	This type of metaphor described depression as whole body or spirit, or a particular part of the body, such as the head or heart feel heavy.	1.My feelings, far from improving, became very heavy. 2.I saw the whole world in black.
Container	This type of metaphor describes depression as a kind of conceptual bounded space, or the patient himself is some kind of conceptual container, the patient himself feels bound, oppressed, or abstractly unable to move	1.I locked myself inside myself. 2.We can't run away from ourselves.
War	This type of metaphor describes depression as an antagonistic force, similar to an opponent or enemy in battle.	1.I'm constantly fighting with my depression. 2.It s a war inside my head.
Descent	This type of metaphor describes depression as a process of conceptual downward progression, or the body, mind, or environment at some low point	1.I'm sinking deeper into despair. 2.I feel like I'm in hell.
Living Organism	This type of metaphor describes depression as something living, or something that has biological behavior.	1.It keeps growing inside me. 2.It' s like a parasite feeding on my joy.
Other	A sentence containing a metaphor related to depression, such as comparing depression to an abstract thing, a conceptual journey, or the patient likening themselves to something, and other sentences that do not belong to the other six metaphorical source domains.	1.My journey through depression. 2.I feel like little waves of panic wash through me all day.

Table 8: The proportion of unavailable responses (%) of the models in different tasks. '—' indicates no unavailable responses.

	Depression detection	Depression level recognition	Metaphor detection
Model	False Ratio	False Ratio	False Ratio
LLaMA3.1-ZS	69.68	60.72	—
LLaMA3.1-FS	79.31	95.71	—
LLaMA3.1-FT-ZS	—	44.42	—
Qwen2.5-ZS	—	—	—
Qwen2.5-FS	—	6.00	—
Qwen2.5-FT-ZS	—	1.42	—
gemma-ZS	7.81	34.54	1.27
gemma-FS	—	63.23	—
mistral-ZS	—	—	15.05
mistral-FS	—	5.13	36.52
llama3.1-70B-ZS	1.60	3.99	—
llama3.1-70B-FS	—	—	—
GPT-4o-ZS	—	—	—